BikeBalance

Bicycle rebalancing for a small Bluebikes station network

PROBLEM

An empty station blocks a pickup.
A full station blocks a return.

For bike-share operations planners deciding where to move bikes.

Where should bikes move when we can only move a limited number?



Photo: City of Cambridge, Harvard Square (2022)

Yunji Seo / 1.125 / October 25, 2026

Proposed solution

A planning tool for adjusting inputs and comparing rebalancing options.

1 Choose the inputs

Load data, adjust station targets, and set a limit on bikes moved.

2 Review a transfer plan

See how many bikes to move and where on a station map and transfer list.

3 Compare strategies

Compare no action, a greedy strategy, and optimization under the same limits.

Example scenario

Station	Before	Target	After
A	12	8	8
B	2	8	8
C	10	8	8

Move 4 bikes from A to B and 2 from C to B.

Move 6 bikes to meet all three station targets.

Hypothetical example: 16 usable docks per station and a 6-bike movement limit.

Technical approach

Data and software

Bluebikes GBFS: station locations, available bikes, docks, and timestamps.

Python + pandas: check and combine data.
SciPy MILP: optimize bike transfers.
Streamlit + Plotly: controls and station map.

Separate data processing, optimization, and the interface for independent testing.

Runs locally with public data and open-source libraries. No paid API or cloud service planned.

Sources: Bluebikes System Data / SciPy milp documentation

Optimization model

1. **Minimize total absolute gaps from targets.**
2. **For equally balanced plans, minimize bike-kilometers.**

Final stock = current stock + inflow - outflow

Move whole bikes from surplus to deficit.
Preserve total bikes and stay within capacity.
Keep total bikes moved within the limit.

Bike-km = bikes moved × straight-line distance.
This measures distance, not actual truck costs.

Scope and risks

Project scope

15-20 selected Cambridge stations.
One snapshot and one rebalancing decision.
Targets start at 50% of usable capacity.
Users can edit targets and save scenarios.

Model assumptions

No customer trips during modeled transfers.
Treat all usable bikes alike.
Exclude broken bikes and unavailable docks.

Beyond this project: demand forecasting, truck routing, citywide planning, and live dispatch.

Risks and responses

Outdated or inconsistent data

Show timestamps, check for invalid records, and provide a saved snapshot as a backup.

Targets may not match demand

Compare different targets. Better balance alone does not prove better service.

Some targets cannot be reached

Show unmet targets and remaining imbalance.
Report whether the plan is optimal.

Plan and success criteria

Milestones

Proposal and feedback

Confirm data access and review the model.
Refine the scope based on feedback.

Midpoint check-in

Working data loader and planning algorithms.
An interface to generate and compare plans.

Final delivery

Test unusual cases and improve usability.
Document setup and prepare the final demo.

Success criteria

Valid plans
Respect available bikes, usable dock space, and the limit on bikes moved.

Strategy comparison
Compare with no action and a greedy strategy.
Check target gaps first, then bike-km for ties.

Usability and reproducibility
Users can change inputs and understand results.
Saved inputs produce consistent results.